

Additional guidance for 1(a)(ii)**Method 3**[Assuming that] $r = 0.7 \, \Omega$:Total resistance = $7.30 \, \Omega$

$$\therefore \text{Current} = \frac{4.80[\text{V}]}{7.30[\Omega]} = 0.658 \text{ A} \quad (1)$$

$$\begin{aligned} \text{Terminal pd, } V = E - Ir &= 4.80 - 0.685 \times 0.7 \\ &= 4.34 \text{ V, which is approximately as expected} \end{aligned} \quad (1)$$

So the initial assumption is correct

Method 4[Assuming that] $r = 0.7 \, \Omega$:[Total resistance = $7.30 \, \Omega$]

Using the potential divider equation:

$$V = E \times \frac{R}{R+r} = 4.80[\text{V}] \times \frac{6.60[\Omega]}{7.30[\Omega]} \text{ equation used correctly} \quad (1)$$

$$= 4.34 \text{ V, which is approximately as expected} \quad (1)$$

So the initial assumption is correct